AIVORA 2K26 HACKATHON · AI PRODUCTIVITY & AUTOMATION

NagrikSetu

Find the benefits. Fix the paperwork. Reach the right counter.

NagrikSetu helps citizens discover municipal welfare schemes, verify documents, and prepare for applications.

Team Whitebox | Municipal Welfare · India | Multi-lingual | Working MVP

PROBLEM STATEMENT

Allocated on paper, lost in process

Across India's **4,400+ Urban Local Bodies (ULBs)**, an estimated **₹15,000+ Crore** of budgeted social welfare fails to reach eligible citizens every year. The money is sanctioned. The eligibility is real. **What breaks is everything between the two.**

National aggregators such as `myScheme.gov.in` index central and state schemes well — they cover **0 of the 4,400+ ULBs**. Municipal welfare is published instead one city at a time, as scanned regional-language circulars on individual corporation portals.

₹233 Cr

surrendered unspent in the cited PMC FY24 figures

61.35% lapse rate — the same finding, expressed as a rate: **₹380 Cr allocated → ₹147 Cr actually spent → ₹233 Cr returned**.

₹380 Cr	₹147 Cr	61.35%
Budgeted, PMC social welfare FY24	Disbursed to beneficiaries	Lapse rate on the same allocation

60% unspent

in the stated municipal sample (₹400 Cr across PMC, BMC, BBMP FY23–24)

1 in 3 rejected

applications discarded for clerical defects — not ineligibility

4–8 repeat visits

to a ward counter before a single claim is settled

Source: PMC Municipal Corporation Annual Accounts FY23–24; the ₹400 Cr sample reconciles Budget Estimates against actual beneficiary disbursement (PMC, BMC, BBMP). The wider national shortfall is cited as an **indicative estimate only**, not a measured total.

WHERE THE PROCESS BREAKS

Four failure points — and the working days they consume

01 · DIGITALLY BURIED

Guidelines sit in scanned Marathi and Hindi PDFs with no machine-readable index. A citizen cannot search for what they do not know exists.

02 · ARBITRARY FILING WINDOWS

Schemes open for **15–45 days** at unannounced intervals. A Tehsildar income certificate alone takes **30 days** — papers cannot be ready in time.

03 · FRONTLINE CLERICAL REJECTION

Clerks default to rejecting a single-letter name variation or a post-merger IFSC code. **18–35%** of applications are lost to paperwork defects.

04 · COUNTER OPACITY & MIDDLEMEN

Unwritten rules force **4–8 visits** and **₹400/day** in lost wages — or a broker takes **30–40%** of the grant.

ILLUSTRATIVE SCENARIO — NOT A REAL USER

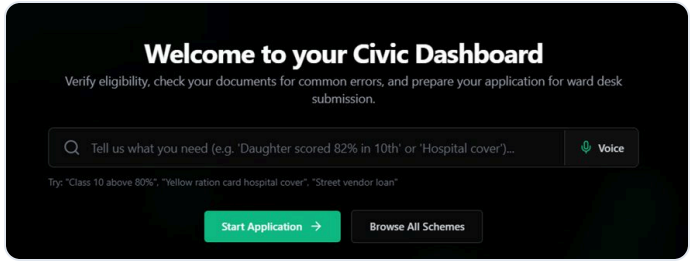
A citizen finds a relevant welfare scheme, but a minor document mismatch — a name spelling variation between Aadhaar and the bank passbook — results in another trip to the ward office.



THE SOLUTION

One guided session instead of a multi-week manual errand

1 Describe need



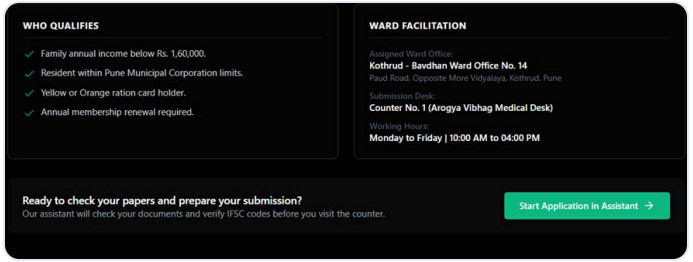
Typed or spoken, in their own language.

2 Find schemes



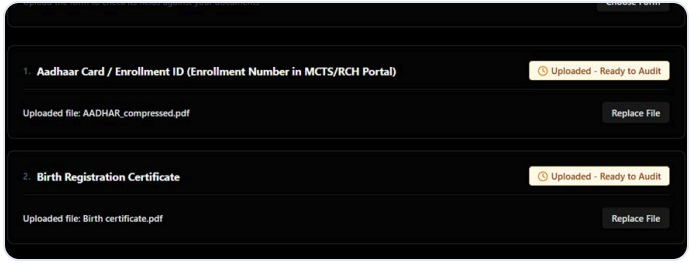
Request becomes a structured search object.

3 Verify eligibility evidence



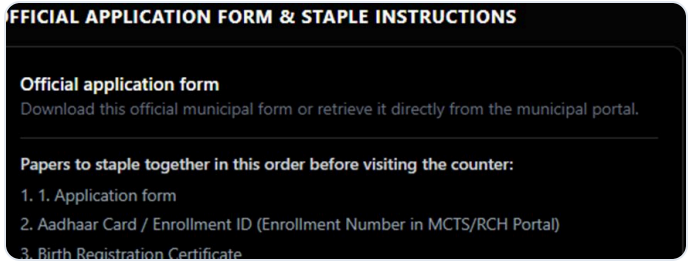
Every match quotes the official lines it relied on.

4 Audit documents



Fields read and defects flagged before the visit.

5 Prepare for submission

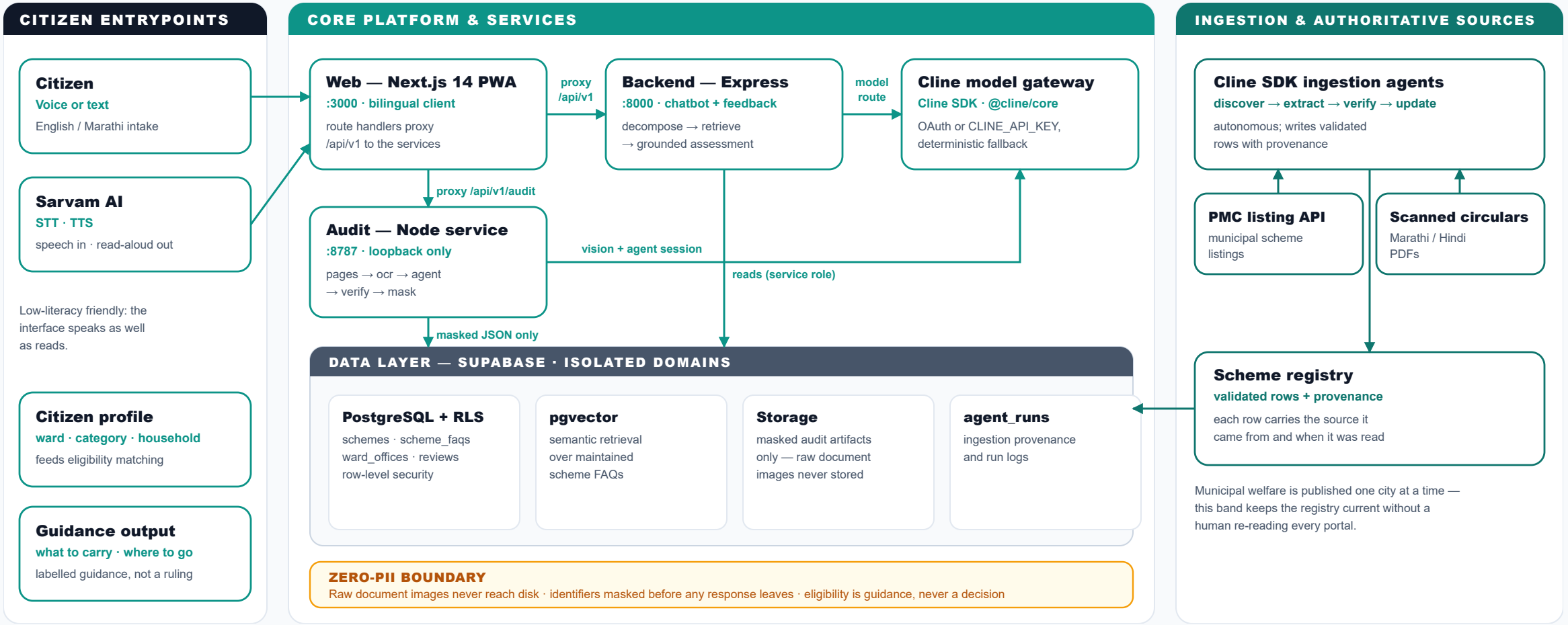


Where to go, what to carry, by when.

WHAT GETS AUTOMATED

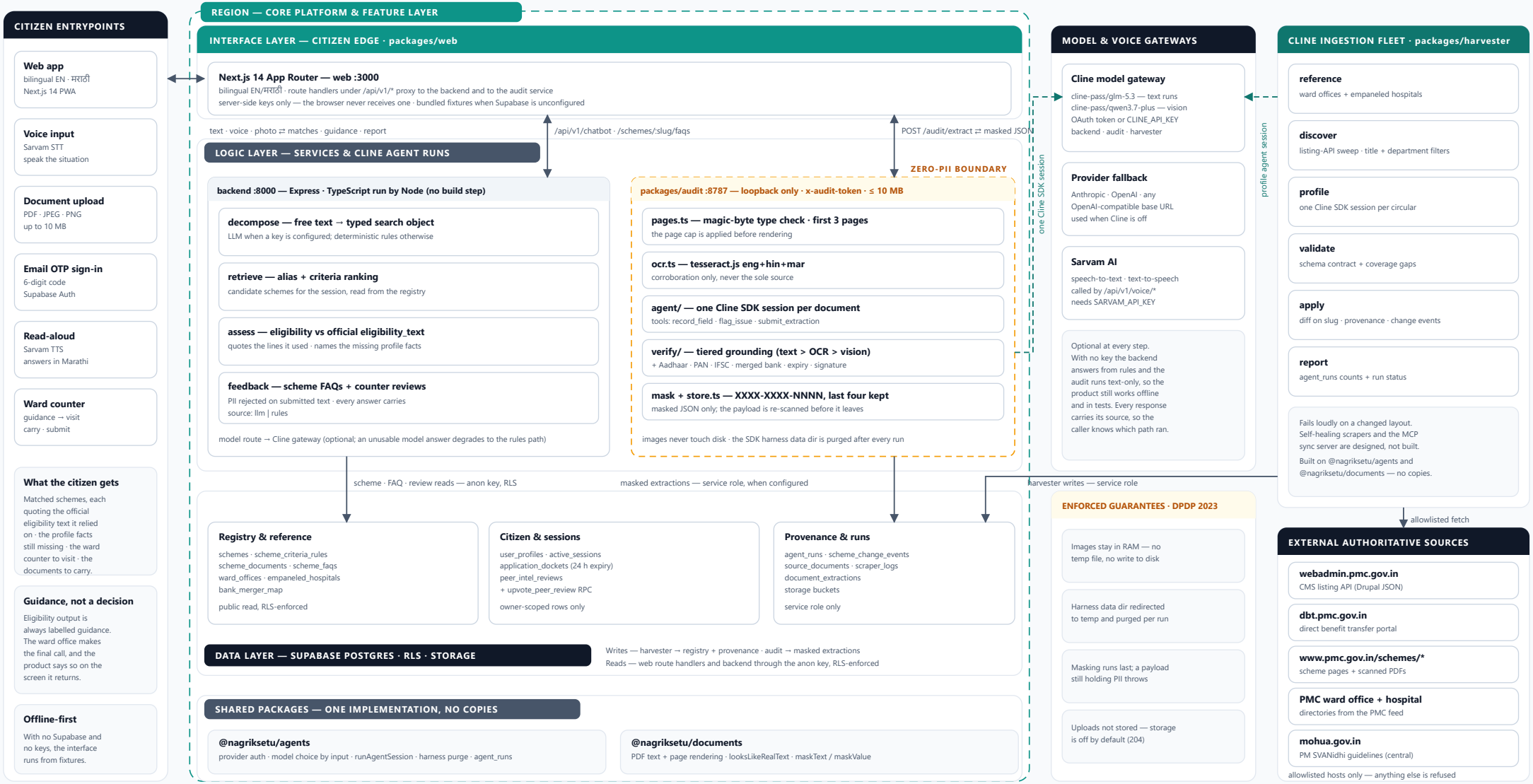
Discovery and indexing of circulars · reading dense regional-language policy PDFs · generating the document checklist · catching typos and invalid paperwork before submission · keeping scheme information current.

How the pieces work together



Three services and four shared packages, all Node.js and TypeScript. Every model call is wrapped in a deterministic fallback; eligibility output is guidance only, never a decision.

SYSTEM ARCHITECTURE



UNDER THE HOOD

Document audit — bytes in, masked JSON out

A. Document processing

Upload → **Extract** → **Validate** → **Mask** → **Re-scan** → **Return result**

Implementation labels: `pages.ts` magic-byte check → `ocr.ts` corroboration → one Cline SDK agent session (`record_field` · `flag_issue` · `submit_extraction`) → `verify/` grounding → `verify/mask` → response re-scan.

B. Privacy safeguards

- **Masking of sensitive identifiers** — Aadhaar `XXXX-XXXX-NNNN`, account and PAN reduced to last four
- **Temporary handling of document images** — raw images never reach disk; the agent harness store is redirected to a volatile directory and purged after every run
- **Re-scanning responses** for unmasked identifiers — a payload that still leaks is never returned
- Only statuses, labels and reasons leave the pipeline — never a field value

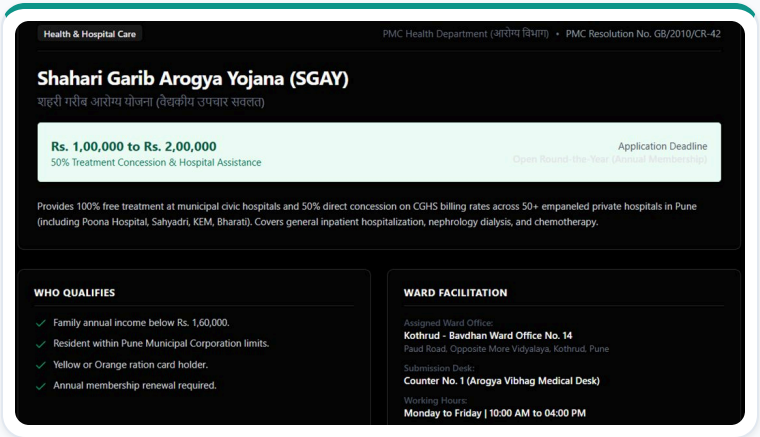
C. Deterministic validation — format checks

- Aadhaar **checksum** validation
- PAN and IFSC **format** checks
- Expiry-date checks
- Missing-signature and missing-seal detection

NAGRIKSETU IN ACTION

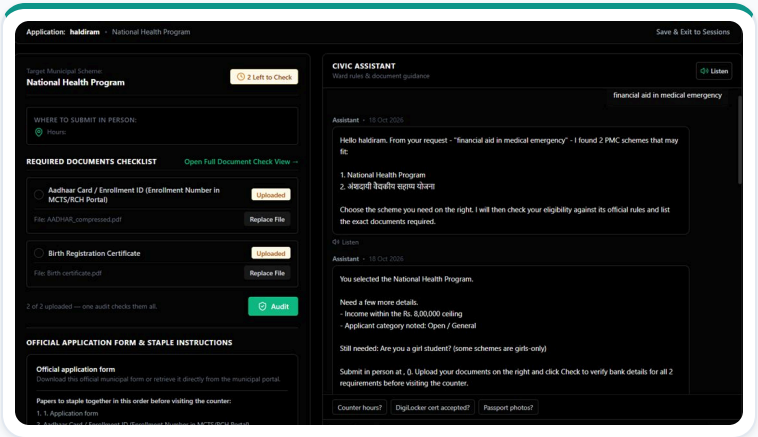
What the citizen does — and what the system returns

CITIZEN EXPERIENCE	SYSTEM OUTPUT
Citizen describes their need	Relevant scheme results, ranked
Scheme is selected	Official eligibility evidence — quoted, with missing facts named
Documents are uploaded	Audit report with extracted fields and detected defects
Citizen reviews next steps	Checklist and office guidance — where to go, what to carry



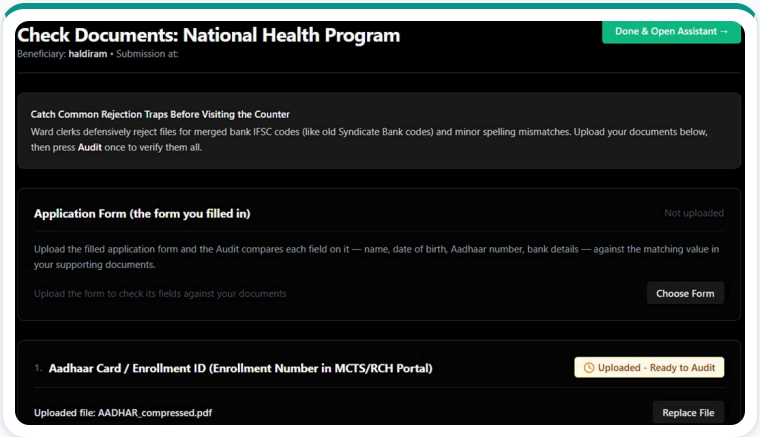
Grounded eligibility evidence

Every scheme states who qualifies — and the ward office and counter to visit.



An assistant that asks for what is missing

Matched schemes, the profile facts still needed, and counter guidance in one thread.



Pre-submission document audit

Every required paper is read and checked before the citizen visits the office.

IMPACT

Expected benefits — and how we will measure them

FOR CITIZENS

- Less time spent searching for schemes
- Better preparation before visiting the office
- Earlier detection of document defects
- Reduced dependence on brokers for interpretation

FOR MUNICIPAL OFFICES

- Fewer avoidable documentation errors
- Potentially less repeat footfall at the counter
- Better access to scheme information for applicants
- Higher absorption of allocated welfare budgets

HOW WE WILL MEASURE IMPACT

- Time to find a relevant scheme
- Accuracy of eligibility matching against verified examples
- Number of document defects detected
- User-reported repeat visits
- % of recommendations supported by current official sources

NagrikSetu does not create new welfare. It removes the process friction that keeps existing welfare from reaching the people it was budgeted for.